

A reporting platform that can explain and repair itself.

From changing documents to governed facts, deterministic workbooks, canonical publication, bounded self-healing, read-only operational experiences, and rollback-safe one-writer delivery.

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4	2	1	0
report families	visible sheets	active writer	retry PUTs after ambiguity

PUBLIC-SAFE SCOPE

This independently written teaching artifact uses generalized architecture and synthetic examples. It excludes production source, history, customer or employee data, credentials, private identifiers, exact topology, ACLs, and live report templates.

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01 / EXECUTIVE SUMMARY

The hard part was not generating a workbook once.

It was proving that late inputs, changing source documents, lost responses, repairs, and releases could not silently create stale or duplicate canonical outputs.

Problem

A healthy component could coexist with a missing or stale report. Byte-level change did not always mean business change, and naive retry could write twice.

Objective

Make every expected output explainable, recover eligible technical gaps, preserve human and security holds, and keep exactly one publisher.

Approach

Join governed intake, semantic identity, deterministic rendering, conditional same-item publication, exact GET readback, and durable repair.

Result

Four report families shared a naturally verified reliability domain with one writer, exact rollback, and equivalent-cycle zero-write evidence.

OWNERSHIP

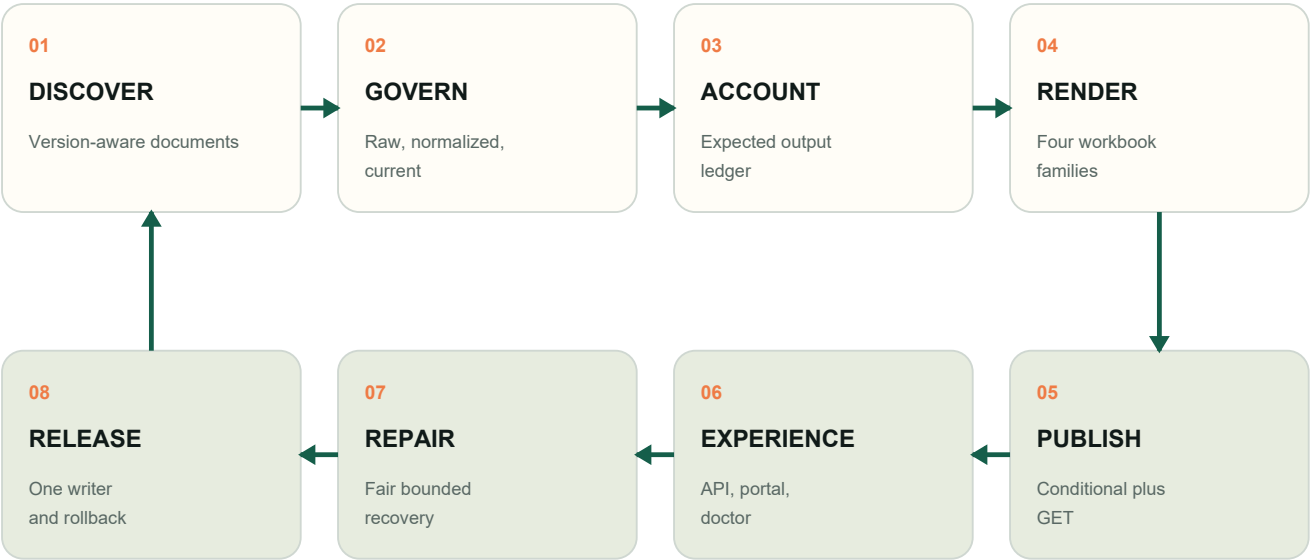
Thomas Ryan set product goals, business definitions, safety invariants, release priorities, acceptance criteria, and owner-only decisions. He materially owned production diagnosis, implementation direction, regression strategy, release and rollback decisions, and evidence-driven handoff. Codex-assisted execution accelerated implementation and documentation under that direction.

02 / ARCHITECTURE

A closed source-to-user reliability loop.

Capabilities are shown at the control-flow level so the whole system is understandable without publishing production coordinates or code.

GOVERNED SOURCE-TO-USER LOOP



AUTHORITATIVE SOURCE	WRITE BOUNDARY	USER SURFACES
Original evidence remains upstream authority. Generated workbooks never become repair sources.	One worker conditionally advances canonical items. Ambiguous writes settle through GET.	API, portal, assistant summaries, and report doctor remain read/interaction planes.

03 / DATA AND PRODUCTS

Governed meaning before presentation.

The reporting product is only trustworthy when source lineage, business semantics, stored workbook state, and visual usability agree.

SURFACE	CONTROL
Discovery	Bounded, version-aware intake separates live changes from historical reconciliation.
Immutable source	Original evidence is retained and remains the only valid repair authority.
Normalized facts	Field-scoped validation lets independent facts progress while unresolved values stay missing.
Governed material	Capture-only metadata is excluded from idempotency; real business changes remain material.
Four families	Shift Draft, Daily, Weekly, and Weekly Downtime share reliability controls but keep family semantics.
Workbook contract	Two visible sheets, formulas, stored values, lineage, layout, and missing-versus-zero behavior are certified.

WHY RAW PACKAGE INSPECTION MATTERS

A spreadsheet client can recalculate formulas and mask a file that lacks required stored values. Certification therefore inspects rendered appearance and the raw workbook package before publication.

04 / PUBLICATION PROTOCOL

One canonical item. One conditional write. One exact readback.

The write protocol treats uncertainty as a reason to reduce authority, never as permission to repeat a mutation.

01	Prepare Retain source, material, and candidate identity before mutation.
02	Compare Equivalent governed material becomes an exact no-op.
03	Write A material candidate advances the approved item conditionally.
04	Read GET verifies exact item, version, ETag, content, and access state.
05	Settle An ambiguous response is resolved by readback without a second PUT.

**AMBIGUITY DOES NOT
INCREASE WRITE
AUTHORITY**

Candidate identity plus exact GET readback is the recovery path. Replay PUT is explicitly excluded.

05 / SELF-HEALING

Repair what is technical. Protect what is authoritative.

The loop is durable, fair, bounded, restart-safe, and deliberately unable to invent business facts or rewrite terminal evidence.

OBSERVE	CLASSIFY	PREPARE	REUSE	SETTLE
Compare expected and actual output state.	Ready, waiting, repairable, protected, refused, or terminal.	Discover and normalize sources outside the critical writer lease.	Use the existing family renderer, publisher, queue, and checkpoints.	Exact readback, no-op, bounded retry, or protected closure.

Eligible automatic convergence
Late qualified source discovery; missing technical output; stale readback; interrupted repair state; materially changed canonical output.

Protected from automation
Unknown identity or lineage; missing business evidence; access or ownership conflict; terminal output; ambiguous rollback; uncertain material semantics.

A SELF-HEALING LOOP NEEDS AN EXPECTATION MODEL

Monitoring only existing publications cannot detect a report identity that was never created. Durable source-to-output accounting makes absence observable and lets family-specific recovery adapters act without adding another scheduler, queue, or writer.

06 / OPERATIONAL EXPERIENCES

Expose state without widening write authority.

The platform includes user and operator surfaces, but only the reporting worker may mutate canonical outputs.

Governed API

Read-oriented access to report state and normalized summaries under explicit contracts.

Portal experience

A discoverable, operations-facing surface for reports, evidence, and governed interaction.

Report-health doctor

Explains fresh, pending, repairable, protected, and terminal states without proof writes.

Assistant summaries

Client-independent stored summaries bind conversational answers to governed report evidence.

Document knowledge

Prepared behind privacy, retention, access, and ownership gates; not labeled generally active.

Interaction telemetry

Privacy-safe citation interaction is implemented and tested with deployment state stated separately.

07 / SECURITY AND RELEASE

Fail closed, ship without planned outage, restore exactly.

Reliability includes identity, permissions, secrets, lineage, presentation, topology, and rollback - not only application health.

Access	Least privilege, explicit identity, protected ownership, no broad edit convenience.
Data	Original-source authority, field-scoped meaning, missing distinct from zero.
Publication	Conditional same-item write, exact readback, no second PUT after ambiguity.
Shadow	Successor prepares and decides in production shape with output suppressed.
Handoff	The current writer drains before the immutable successor activates.
Rollback	Material regression restores the exact certified predecessor.

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A FAILED-RESTORED RELEASE IS EVIDENCE

When a successor fails a material gate and the exact prior release resumes with one writer and preserved service, the result proves containment and rollback. It does not convert the failed successor into a success.

08 / EVIDENCE

Every claim carries an evidence grade.

Local tests, integration, raw artifact checks, shadows, exact readback, failed-restored releases, and natural operation are reported separately.

Focused tests	Bounded rules and synthetic regressions
Integration	Cross-component state and publication contracts
Raw artifact	Sheets, stored values, formulas, lineage, package state
Zero-output shadow	Production-shaped preparation and decisions without mutation
Exact readback	Canonical item identity, version, ETag, content, and access
Natural schedule	Real cadence produced the intended write, settlement, or no-op

VERIFIED OUTCOMES

- Deterministic two-sheet products with client-independent stored values.
- Independent expected-output state capable of detecting a missing identity.
- Qualified late evidence advanced the canonical item at most once.
- Accepted writes settled through exact GET readback without retry PUT.
- Equivalent natural replay produced an immediate zero-write no-op.
- Immutable handoff and exact rollback preserved one writer.

09 / CHRONOLOGY AND LESSONS

From source foundation to natural no-op closure.

The public record contains 29 dated milestones. The engagement began in May; day-level public evidence starts in late June and is not invented where records are absent.

JUL 01	Authoritative final-shift source foundation	AUG 01	Recurring orchestration activated
JUL 23	Integrated data-platform baseline	AUG 09	Accepted-write GET recovery
AUG 14	Closed-loop self-healing and lease isolation	AUG 17	Client-independent workbook correctness
AUG 20	Expected-output accounting and security hardening	AUG 22	Natural governed change and exact no-op

LASTING ENGINEERING LESSONS

- Idempotency needs governed semantic identity, not only a file hash.
- Ambiguous network responses should reduce write authority.
- Self-healing is safe only when terminal and human-owned holds remain protected.
- Raw artifacts matter when a client can recalculate and hide defects.
- Optional work should not hold the lease protecting user-facing output.
- Failed-restored releases and zero-write no-ops are first-class evidence.

CONTACT

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